FindEase – Intelligent Lost & Found Portal

A Mini Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINNERING

Or

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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DECLARATION

The Mini Project Report entitled "**FindEase – Intelligent Lost & Found Portal** "is a record of Bonafide work of **K. Bhavya-2420030394 ,M. Pranavi-2420090110, S.Saswitha-2420090067.**submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering (or) Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled "**FindEase – Intelligent Lost & Found Portal"** is a bonafide work done and submitted by **K. Bhavya-2420030394 ,M. Pranavi-2420090110, S.Saswitha-2420090067.** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Course Co-ordinator **Signature of HOD**

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1. ABSTRACT

ABSTRACT

Lost and misplaced personal belongings are a common concern in educational institutions, where students and staff move frequently between classrooms, laboratories, libraries, cafeterias, and other campus facilities. Traditional recovery methods, such as physical notice boards, paper registers, and informal messaging groups, can be difficult to organize. They may also expose personal contact details and reveal item characteristics that could be misused by people making false claims.

FindEase is a web-based campus lost-and-found management system designed to make item reporting and recovery more organized while reducing unnecessary disclosure of personal information. The platform allows users to submit lost-item and found-item reports with relevant details such as title, category, description, and location. Category- and keyword-based matching helps users identify potentially related reports. For found items, the finder can configure a private verification question so a claimant must provide relevant knowledge before the claim is reviewed. The finder can then accept or reject the claim. After acceptance, users can coordinate the handover through an item-specific in-app messaging workflow without directly sharing phone numbers or email addresses through the chat.

The system follows a decoupled Service-Oriented Architecture approach, with a React.js and Vite frontend, a Java Spring Boot REST API backend, and a MySQL relational database accessed using Spring Data JPA and Hibernate. The project demonstrates how modular web technologies can support structured reporting, claim review, and privacy-conscious communication in a campus setting.

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2. INTRODUCTION

2.1 Background

Students and staff carry personal belongings such as identity cards, keys, wallets, notebooks, mobile phones, and electronic accessories across a campus every day. When an item is misplaced, the owner may not know where to report it or how to reach the person who found it. A centralized digital platform can provide a structured way to publish reports, search for possible matches, and coordinate recovery.

2.2 Existing Challenges

Scattered information: Lost-item reports may be distributed across messaging groups, notice boards, or separate offices.

Privacy concerns: Public posts may expose phone numbers, email addresses, or other personal information.

False claims: If complete item details are visible publicly, a person may attempt to claim an item without being its rightful owner.

Limited traceability: Informal methods may not provide a consistent record of report status, claim review, or recovery communication.

2.3 Proposed System

FindEase provides a single web interface for reporting lost and found items. Users can view relevant listings and identify possible matches through category and title-keyword matching. A finder can define a private challenge question for a found item, and a claimant can submit an answer as evidence of ownership. The finder reviews the claim and records an acceptance or rejection decision. For accepted claims, the platform provides an item-scoped chat workflow using masked participant roles, allowing users to coordinate a safe handover without needing to disclose direct contact details in the conversation.

2.4 Scope of the Project

The current project scope covers web-based item reporting, item listing and matching, ownership-claim submission, claim review, item status updates, and in-app messaging associated with an item. It is intended as a campus-level software solution. Features such as institutional single sign-on, push notifications, image-based AI matching, smart lockers, and cross-campus federation are proposed as future enhancements rather than assumed current capabilities.

2.5 Organization of the Report

This report presents the background and existing approaches, defines the problem and objectives, lists system requirements, describes the architecture and implementation, provides a framework for presenting actual application results, and concludes with limitations and future development opportunities.

3. LITERATURE SURVEY

3.1 Physical Notice Boards and Security Registers

Physical notice boards and paper registers provide a straightforward way to record found property. However, they usually require people to visit a particular location, and information may not be searchable remotely. Manual processes can also make it difficult to keep item status current or maintain a consistent history of claim decisions.

3.2 Messaging and Social Media Groups

Messaging groups can spread information quickly within a campus community. However, reports may be buried under unrelated messages, and users may be encouraged to publish personal contact information or detailed descriptions. These channels generally do not provide a dedicated workflow for submitting and reviewing ownership claims.

3.3 Online Lost-and-Found Platforms

Digital lost-and-found platforms offer structured listings and make it easier to search reports across a venue or community. Their suitability depends on the visibility of item details, the available verification process, and the communication features. FindEase emphasizes private challenge-based claim review and item-scoped messaging as part of its campus-focused workflow.

3.4Bluetooth Tracking Devices

Bluetooth tracking products can help locate compatible tagged belongings. However, they require a compatible tracking device to be purchased and associated with an item, and they do not cover every everyday object. A web-based reporting system can complement such devices by supporting items that have no tracker.

3.5Institutional Lost-Property Services

Some institutions manage lost property through campus security or administrative offices. These services provide an official point of contact, but a manual process may require staff to coordinate reports and claimants. FindEase aims to provide a structured online workflow that can support, rather than necessarily replace, institutional procedures.

3.6 Comparative Summary

Approach	Common limitation	FindEase approach
Physical register / notice board	Limited remote access and searchability	Centralized digital reports
Messaging groups	Posts can be buried; contact details may be exposed	Structured listings and in-app communication
General online listings	Verification and privacy controls vary	Private challenge question and claim review
Bluetooth trackers	Requires compatible hardware	Software-based reporting for ordinary belongings
Manual institutional process	May depend on staff coordination	Online item and claim workflow

4. Problem Statement and Objectives

4.1 Problem Statement

Traditional lost-and-found systems in educational institutions rely on physical notice boards, manual registers, and informal communication platforms such as WhatsApp groups. These methods often make it difficult for students to report lost belongings, identify matching found items, and recover their property efficiently. Sharing complete item descriptions and personal contact information publicly may also lead to privacy risks and fraudulent ownership claims.

Therefore, there is a need for a centralized, secure, and user-friendly web application that simplifies the lost-and-found process, supports ownership verification, and enables users to communicate without unnecessarily disclosing their personal information.

FindEase addresses these challenges by providing a digital platform for reporting lost and found items, identifying potential matches, verifying ownership through private challenge questions, and coordinating item recovery through in-app communication.

4.2 Problem Description

The existing lost-and-found process has several limitations:

- Lack of centralization: Information about lost and found items is scattered across different platforms and physical locations.
- Privacy issues: Sharing phone numbers, email addresses, and personal details publicly may expose users to unwanted contact.
- Fraudulent claims: Unauthorized individuals may attempt to claim items by using publicly available descriptions or photographs.
- Time-consuming recovery: Owners may find it difficult to locate relevant reports and contact the person who found their belongings.
- Lack of ownership verification: Traditional methods often do not provide a structured process for verifying whether a claimant is the rightful owner.

4.3 Objectives

1. Item Reporting: Allow users to report lost and found items.
2. Automatic Matching: Identify potential matches using item categories and keywords.
3. Ownership Verification: Use private challenge questions to help verify claims.
4. Secure Communication: Enable in-app messaging without directly sharing personal contact details.
5. Claim Management: Allow finders to accept or reject ownership claims.
6. Efficient Recovery: Simplify the process of finding and returning lost belongings on campus.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or equivalent
- RAM: 4 GB
- Storage: 10 GB free space

Software Requirements

- OS: Windows/Linux
- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Java, Spring Boot
- Database: MySQL
- Tool: VS Code
- Browser: Google Chrome

6. IMPLEMENTATION

The FindEase system is developed using a three-tier architecture consisting of frontend, backend, and database.

1. Frontend Implementation

The frontend is developed using React.js and Vite. It provides a user-friendly interface where users can register, report lost or found items, view item details, and submit claims.

2. Backend Implementation

The backend is developed using Java and Spring Boot. It handles user requests, manages item reports, processes ownership claims, and supports secure communication between users through REST APIs.

3. Database Implementation

MySQL is used to store user details, lost and found item information, claim requests, and messages. It helps manage and retrieve data efficiently.

4. Item Reporting

Users can submit details of lost or found items, including item names, categories, descriptions, and other relevant information.

5. Item Matching

The system identifies possible matches between lost and found items using their categories and keywords, helping users find their belongings easily.

6. Claim Verification

When a user claims an item, a private challenge question is used to help verify ownership. The finder can accept or reject the claim.

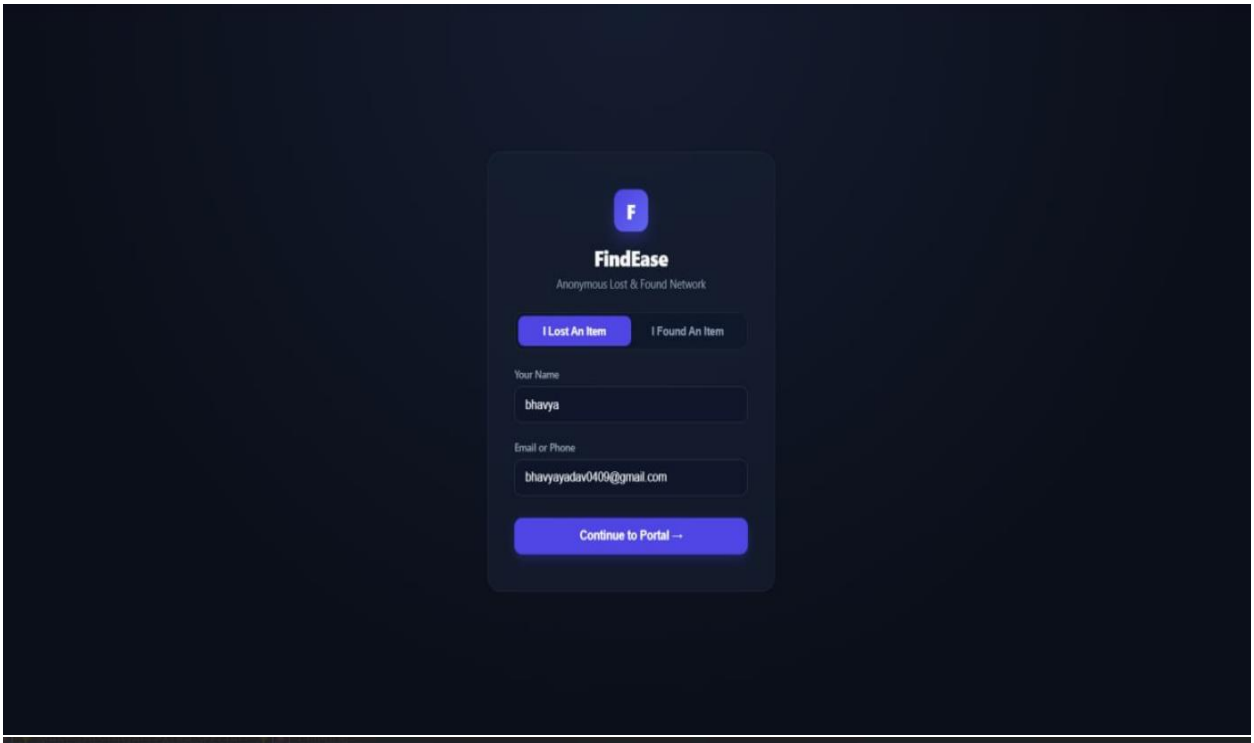
7. Secure Messaging

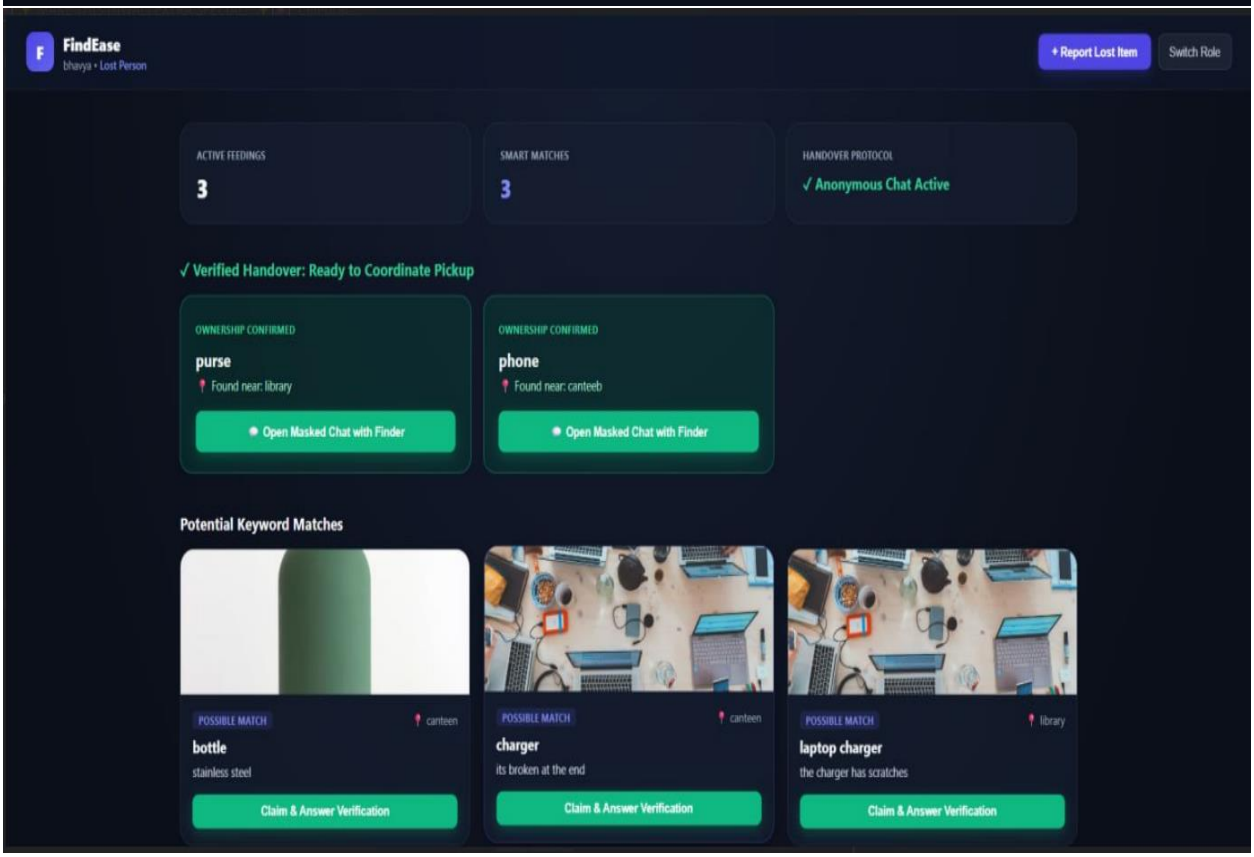
FindEase provides item-specific in-app messaging so that owners and finders can communicate without directly sharing their personal contact details.

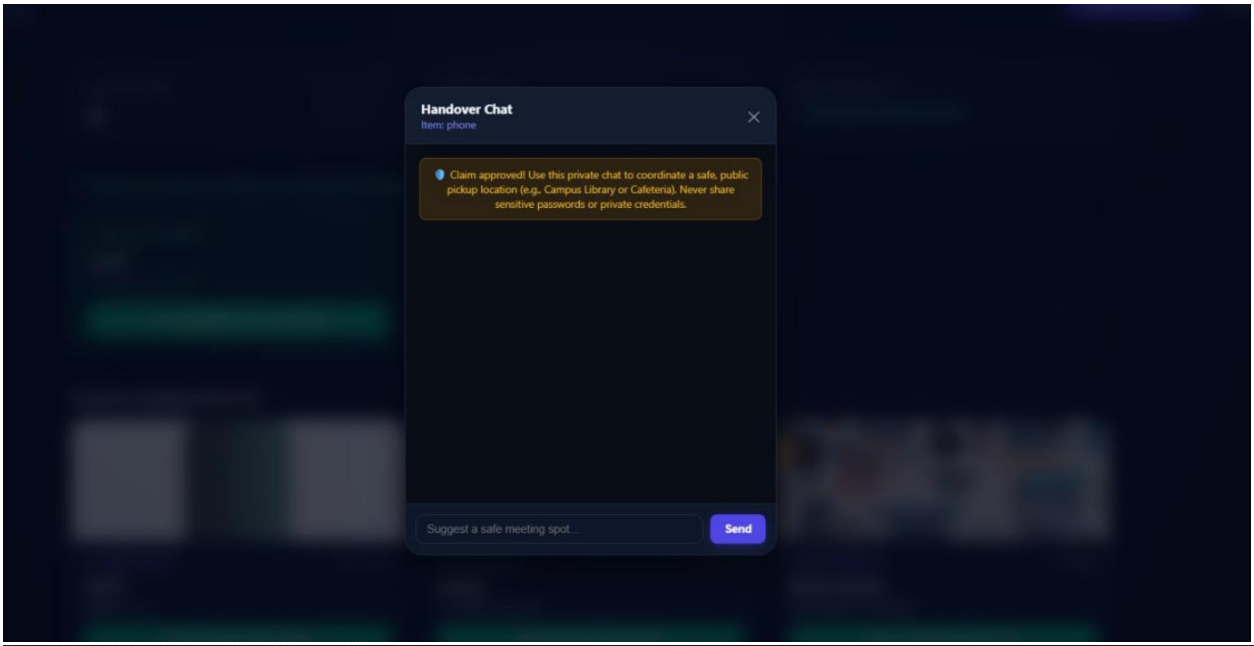
8. Overall Working

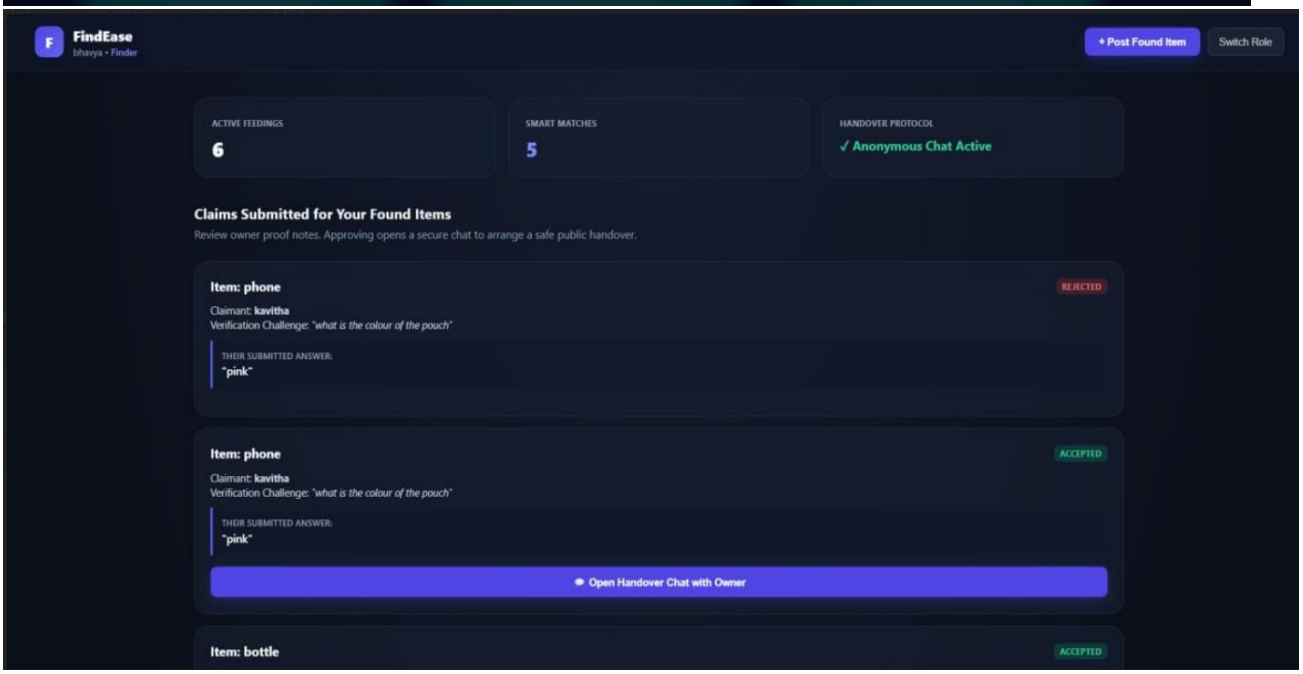
The system connects users, manages item reports, identifies potential matches, verifies ownership claims, and supports communication. This makes the campus lost-and-found process more organized, secure, and efficient.

7. RESULTS & DISCUSSION









This section should present evidence from the running FindEase application. Insert screenshots captured from your actual implementation and write a short explanation under each screenshot. Do not report performance figures, recovery rates, or security guarantees unless they have been measured or verified.

8. **CONCLUSION & FUTURE SCOPE**

Conclusion

FindEase provides an efficient and secure solution for managing lost and found items on campus. It allows users to report missing and found items, identify possible matches, verify ownership through challenge questions, and communicate through in-app messaging. The system reduces manual effort, improves privacy, and makes the process of recovering lost belongings easier and more organized.

Future Scope

FindEase can be improved in several ways to make the system more advanced and useful in the future.

1. Mobile Application: A mobile application can be developed for Android and iOS so users can access the system easily from their smartphones.

2. Image-Based Matching: Image recognition technology can be added to compare uploaded item images and identify similar lost and found objects.

3. Notification System: Email or mobile notifications can be introduced to alert users when a matching item is reported or when a claim request is updated.

4. AI-Based Matching: Artificial intelligence can be used to improve matching accuracy by analyzing item names, descriptions, categories, and images.

5. Location-Based Search: Campus location details can be added to help users identify where an item was lost or found.

6. Multi-College Support: The system can be expanded to support multiple colleges, universities, and other institutions.

7. Improved Security: Additional security features can be introduced to strengthen user authentication, protect personal information, and prevent unauthorized access.

8. Admin Dashboard: An advanced admin dashboard can be developed to monitor reported items, manage users, handle complaints, and maintain the overall system.

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